

Superconductivity of large bipolarons

As the temperature is lowered below a critical value the transport properties of some materials display the distinctive features which taken together define “superflow”. The superflow of electrons and helium atoms describe superconductivity and superfluidity, respectively. With superconductivity electrons flow without resistance and with a vanishing Seebeck coefficient while expelling magnetic flux (the Meissner effect). These features imply that carriers manifest superconductivity when they condense into a collective ground state which precludes their scattering and their orbital paramagnetism. Analogously, helium atoms manifest superfluidity when they condense into a collective ground state which precludes their scattering and their rotational flow.

The mobile species of these collective ground states are bound together by their mutual attraction. The van der Waals (electrostatic) attraction between helium atoms drives their condensation into a liquid. The superfluid- ^4He phase is nested within its liquid phase. A phonon-mediated attraction among the conduction electrons of a metal having energies close to its Fermi level produces the condensate for BCS superconductivity (Bardeen *et al.*, 1957a; 1957b). In particular, the attraction between electrons is mediated by their mutual coherence with phonons. The stabilization of the condensate depends on electron’s coherent response to atoms’ zero-point dynamics.

It has long been conjectured that mobile real-space singlet pairs might form the basis of a superconducting condensate analogous to that for liquid ^4He (Schafroth, 1955). Small- and large-singlet bipolarons constitute real-space singlet pairs associated with displacing the equilibrium positions of surrounding atoms. As discussed in Chapter 10, small polarons are generally severely localized and move incoherently by thermally assisted hopping. As such they are not expected to be a suitable basis for bipolaronic superconductivity (Anderson and Abrahams, 1987; Emin and Hillery, 1989; Chakraverty *et al.*, 1998). By contrast, in analogy with large polarons, large bipolarons are presumed to move coherently.

To obtain a form of superconductivity that is akin to the superfluidity of ^4He , interactions between large bipolarons would have to facilitate their condensing into a liquid. The ground state of this interacting system would also have to remain fluid rather than solidifying. Finally, the excitation spectrum of this fluid ground state would then have to support resistance-less flow and the Meissner effect. A discussion of this scenario begins by addressing the interactions between large bipolarons.

14.1 Interactions between large bipolarons

Large bipolarons, each carrying the electronic charge of two carriers, repel one another via their long-range Coulomb repulsion. These repulsions are modulated by the static dielectric constant ϵ_0 since self-trapped carriers can move no more rapidly than the atoms whose displacements bind them. Thus, the Coulomb repulsion between two bipolarons of electronic charge $2e$ separated by the distance s is written as $(2e)^2/\epsilon_0 s$, where s exceeds a bipolaron's radius R_b (Emin, 1994a; 1994b).

An additional repulsive effect comes into play as two large bipolarons get close enough to one another to begin to merge into a “quadpolaron”. In particular, as illustrated in Fig. 7.1, the Pauli principle dictates that only one singlet pair occupies the lowest non-degenerate level of the self-trapping potential well. The energy associated with promoting two carriers above the self-trapping well's lowest level destabilizes a large quadpolaron with respect to two separate large bipolarons (Emin, 1994b). In other words, large bipolarons experience a short-range repulsion of one another. This repulsion has been modeled as a hard-core repulsion akin to that between ^4He atoms: $\epsilon_S(s) = C_1 \exp(-C_2 s/R_b)$, where s is the separation between two large bipolarons having radii R_b while C_1 and C_2 denote numerical constants (Emin, 1994a; 1994b).

Attractive interactions between large bipolarons result from their self-trapped electronic carriers adjusting to atoms' vibrations. As described in Chapter 4, the adjustment of a self-trapped carrier to vibrations of the associated atoms reduces the related stiffness constants thereby lowering the vibrations' energy. Moreover, the net lowering of the vibration energy produced by a collection of large bipolarons increases as their mutual separations decrease (Emin, 1994a; 1994b). In particular, a second-order perturbation calculation of the energy shift produced by the softening arising from a collection of large bipolarons depends on their mutual separations. This dependence arises as the self-trapped carriers respond coherently to vibrations whose wavelengths exceed the separation between large bipolarons. The change of the zero-point vibration energy induced by two large bipolarons of radius R_b separated by the distance s is roughly $-\hbar\omega(R_b/s)^4$. This energy lowering describes an attractive force between large bipolarons. Here, as in the BCS theory of metals' superconductivity, the ground-state energy is reduced by electrons' coherent

response to zero-point atomic vibrations. In both instances the magnitude of this energy lowering increases as atoms' vibration frequencies are increased.

14.2 Condensation to a large-bipolaronic liquid

Driven by attractive inter-bipolaron interactions, large bipolarons may condense into a liquid phase in an analogous manner to that by which ^4He atoms condense into a liquid phase. The zero-point phonon-mediated attractions between large bipolarons described in Section 14.1 then replace the direct van der Waals (electrostatic) attractions between ^4He atoms. These attractive interactions will overwhelm large-bipolarons' mutual Coulomb repulsions if they are sufficiently suppressed by a material's exceptionally large static dielectric constant. Indeed, as described in Section 7.1, a very large ratio of static-to-optic dielectric constants is required for large-bipolaron formation. Thus, an especially large static dielectric constant is a reasonable presumption in instances in which large bipolarons form. Then, as illustrated in Fig. 14.1, the sum of the three interactions described in the previous section produces a minimum of the net interaction potential V_{int} near where the inter-bipolaron separation s is comparable to twice the large-bipolaron radius R_b .

Establishing a minimum in the interaction potential insures that large bipolarons will condense at sufficiently low temperatures. Calculation of the associated phase diagram proceeds in analogy with that for the condensation of a gas into a van der Waals liquid. The curve in Fig. 14.2 shows the temperature below which a gas of large bipolarons begins condensing into a bipolaron liquid versus the bipolaron concentration (Emin, 1994a; 1994b). This condensation temperature T_c increases with the depth of the attractive potential while the van der Waals repulsion volume is only qualitatively related to that of a large bipolaron (Landau and Lifshitz, 1958). A vertical dashed line is added to Fig. 14.2 to schematically denote the carrier density

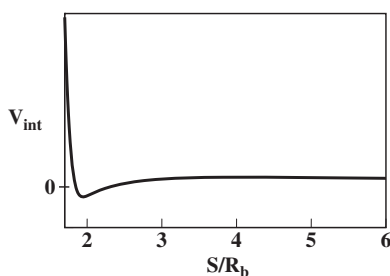


Fig. 14.1 When the static dielectric constant is sufficiently large the interaction potential V_{int} comprising the three components described in Section 14.1 develops a minimum at a separation between large bipolarons s which is near twice the bipolaron radius R_b .

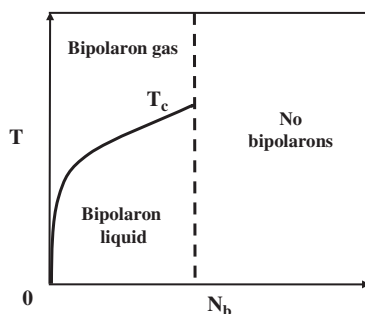


Fig. 14.2 Phonon-mediated attractions between large bipolarons enable them to condense from a gas of nearly independent quasi-particles into a liquid phase. Condensation begins when the temperature T is lowered below the condensation temperature T_c . The condensation temperature rises with the large-bipolaron density n_b . However, large bipolarons cannot form beyond a maximum limiting density depicted by the vertical dashed limit.

beyond which large bipolarons can no longer form. In particular, large bipolarons must be sufficiently sparse to provide enough polarizable medium about each of them to enable self-trapping.

The radius of a Fröhlich large-bipolaron R_b nearly equals that of a Fröhlich large-polaron R_p . In particular, as described in Section 7.1, the contraction of the self-trapped state induced as pairing doubles the depth of the self-trapping potential well offsets the expansion of the state induced by the pair's mutual Coulomb repulsion. The radius of a Fröhlich large polaron is $R_p \equiv (\hbar^2/me^2)[\epsilon_\infty\epsilon_0/(\epsilon_0 - \epsilon_\infty)]$, where m denotes the electronic carrier's effective mass while ϵ_0 and ϵ_∞ , respectively, represent the material's static and optical dielectric constants. Evaluation of this formula indicates that R_p is usually not much larger than an interatomic separation. For instance, (1) $\hbar^2/me^2$ is just the Bohr radius ($\approx 0.5 \text{ \AA}$) if m equals the free-electron mass and (2) $\epsilon_0 \geq 2\epsilon_\infty$ for large bipolaron formation. By contrast, a small bipolaron is one that has collapsed to the smallest size acceptable in a medium composed of discrete atoms and bonds (e.g. just a single atom, ion or bond).

Figure 14.2 only depicts the phase boundary delineating where all large bipolarons exist as a gas and where large bipolarons first condense into a liquid. Moreover, as the temperature is raised large bipolarons in the gas may decompose into large polarons and self-trapped carriers may be liberated thereby spawning free carriers. Furthermore, cooling the liquid can drive some carriers to organize spatially in a manner commensurate with the underlying solid's lattice structure. These carriers may even collapse onto individual ions, atoms or bonds thereby forming regular arrays of small polarons or small bipolarons. That portion of the large-bipolaron liquid which survives as the temperature approaches absolute zero may participate in its superconducting condensate.

14.3 Superconductivity and excitations of a large-bipolaronic liquid

Large-bipolaronic superconductivity, like the superfluidity of ^4He , is attributed to collective behavior of mutually interacting particles whose ground state remains liquid. Liquidity persists in the ^4He ground state because the mutual attractive potential between ^4He atoms is so weak that the amplitude for their zero-point vibrations is of the order of the interatomic separation (Feynman, 1972). A similar situation prevails for a liquid of large bipolarons since their weak ($< \hbar\omega$) mutual attractive potential is comparable to their zero-point kinetic energy $\sim \hbar^2/m_b R_b^2 < \hbar\omega$. One sees that $\hbar^2/m_b R_b^2 \sim (\hbar\omega)^2/E_p < \hbar\omega$ upon recalling from Chapters 3, 4 and 10 that $m_b \sim E_p/\omega^2 R_b^2$ with the large-polaron's binding energy being large enough to insure self-trapping $E_p > \hbar\omega$.

Large-bipolarons' superconductivity is envisioned as analogous to the superfluidity of ^4He . This approach presumes that singlet large bipolarons, like ^4He atoms, behave as bosons. The onset of superconductivity of a large-bipolaronic liquid, like the onset of superfluidity of liquid ^4He , is then taken to occur as the temperature is lowered below the liquid's Bose–Einstein condensation temperature. Below this condensation temperature the excitation-less ground state has a finite occupation even when reduced by the interactions between the condensate's particles (Penrose and Onsager, 1956). The distinctive transport properties which collectively constitute superflow are attributed to this condensate. In particular, the condensate is characterized by resistance-less irrotational flow which yields the Meissner effect when the bosons are charged (Lifshitz and Pitaevskii, 1980).

The condensate's flow will be without resistance if it does not generate excitations. Excitations will not be generated if their creation is incompatible with conservation of energy and momentum. In particular, the Landau condition for resistance-less condensate flow is that its velocity v_c be less than the minimum value of $E(p)/p$, where $E(p)$ is the energy of a real (undamped) condensate excitation of momentum p (Landau and Lifshitz, 1958). The excitations for interacting bosons will be completely undamped if the curvature of the excitation spectrum is negative and momenta are kept below a threshold value (Pitaevskii, 1959; Abrikosov *et al.*, 1963; Lifshitz and Pitaevskii, 1980).

The excitations of the condensate depend on the interactions between its particles. The long-range Coulomb interactions between large bipolarons determine the long-wavelength limit of their excitation energy. As illustrated in Fig. 14.3, the long-wavelength limit of the excitation energy is simply $\hbar\omega_p$, where $\omega_p \equiv (4\pi n_b q^2/\epsilon_0 m_b)^{1/2}$ is the plasma frequency for large bipolarons of density n_b , mass m_b and charge $q = 2e$ in a medium with static dielectric constant ϵ_0 . Large bipolarons' mutual Coulomb repulsions ensure that the Landau condition for resistance-less flow is satisfied at long wavelengths: $E(p)/p \rightarrow \hbar\omega_p/p \rightarrow \infty$ as $p \rightarrow 0$. By contrast, the Landau condition for

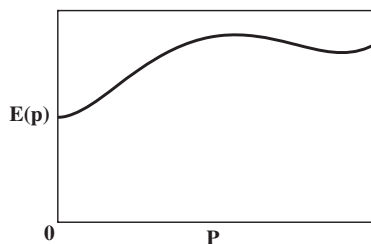


Fig. 14.3 The anticipated momentum dependence of the excitation energy $E(p)$ of a large-bipolaronic liquid is depicted. In the $p \rightarrow 0$ limit $E(p) \rightarrow \hbar\omega_p$ as long-range Coulomb interactions between large bipolarons predominate. The depiction of a ^4He -like “roton” minimum indicates that $E(p)$ may display complex behavior away from its $p \rightarrow 0$ limit.

resistance-less flow in liquid ^4He is satisfied at long wavelengths if the flow velocity v_c is less than the liquid’s sound velocity s_1 : $E(p)/p \rightarrow s_1 p/p = s_1 > v_c$.

The excitation energies of interacting large bipolarons are modest, only comparable to phonon energies. In particular, the hefty large-bipolaron effective mass, the large value of the static dielectric constant required for large-bipolaron formation ($\epsilon_0 \gg \epsilon_\infty > 1$), and the modest large-bipolaron density $n_b \ll R_b^{-3}$ result in $\omega_p < \omega$, the characteristic phonon frequency. In addition, the large-bipolaron’s big effective mass restricts the overall spread of excitation energies to less than the phonon energy $\hbar\omega$.

14.4 Distinctive properties of a large-bipolaronic superconductor

Large-bipolaronic superconductivity involves singlet self-trapped carrier pairs which move very slowly within a medium whose exceptionally displaceable ions generate an especially large static dielectric constant $\epsilon_0 \gg \epsilon_\infty$. By contrast, BCS superconductivity involves fast-moving electronic charge carriers in a conventional metal. Some phenomena highlight these fundamental distinctions.

As described in Section 9.1, incident radiation is absorbed when a self-trapped carrier is excited within or from the potential well which binds it. Such absorptions generally produce phonon-broadened low-energy absorption bands. Absorption bands from exciting self-trapped carriers are produced by polarons and/or bipolarons that are (1) bound to dopants, (2) free of dopants and (3) participants in a superconducting condensate. However, such distinctive absorption bands are not produced by electronic carriers in normal or superconducting states of conventional metals.

(Bi)polarons’ photo-emission occurs when carriers that are photo-excited from their self-trapped states escape from the solid. The Franck–Condon principle ensures that atoms do not move during the excitation process. Thus this photo-emission is similar to that from a doped semiconductor whose carriers are bound in

conventional traps. Moreover, below a bipolaronic superconductor's condensation temperature dispersion of the self-trapping energies will generally narrow as (bi) polarons enter their homogeneous condensate. Distinctively, with bipolaron formation the density of self-trapped states equals the density of self-trapped pairs. Thus, the electrons' chemical potential remains pinned between the energy of the self-trapped carriers and that of the edge of the solid's non-polaronic states. By contrast, the Fermi energy of a non-polaronic material shifts as the states' occupancies are altered.

Positrons injected into a solid can recombine with its electrons. Positron annihilation is evidenced by the emission of gamma rays. The positron annihilation rate depends on the electron density and its homogeneity. In particular, the positron lifetime is sensitive to the presence of localized electrons. Thus, a materials' normal-state positron lifetime should also fall upon introducing self-trapped carriers. However the positron lifetime ought to rise as the temperature is lowered below the condensation temperature where self-trapped carriers merge into the bipolaronic liquid's homogeneous superconducting condensate. By contrast, the positron lifetime of a conventional superconductor is not altered upon passing into the superconducting state.

The shifting of large (bi)polarons' self-trapped charges in response to atomic vibrations reduces their associated stiffness constants. If this carrier-induced softening is strong enough, multi-site self-trapped carriers of radius R_b induce local vibration modes having wavelengths $\leq R_b$ (Emin, 1991b). Thus, large (bi)polarons' presence can introduce short-wavelength vibration modes beyond those of the carrier-free solid. Such "extra" vibration modes have been called "ghost" modes.

All told, large (bi)polarons' self-trapped carriers produce multiple manifestations of localized charge carriers. Indeed, such charge localization can occur in the exceptional ionic materials in which large (bi)polarons can exist but not in conventional metals. Nonetheless, electronic localization by itself does not necessarily imply either self-trapping or large-bipolaron formation. Indeed, electronic localization results from trapping as well as from self-trapping.

Other distinctive large-(bi)polaronic properties directly address electronic transport. As described in Section 10.3, a large (bi)polaron possesses a large effective mass and is weakly scattered by phonons. By contrast, conventional metals' carriers have modest effective masses and are strongly scattered by phonons. In combination these two properties yield moderate mobilities for both large (bi)polarons and for conventional metals' carriers. However, measurements which individually address carriers' effective masses and scattering times distinguish between large (bi)polarons and metals' electronic carriers. In particular, scattering times can be deduced from the frequency dependence of the electrical conductivity. Scattering times for large (bi)polarons should approach atoms' characteristic vibration period. However, scattering times are generally much shorter for metals' charge carriers. Furthermore,

as discussed in Section 10.4, a large (bi)polaron's scattering by acoustic phonons peaks for half-wavelengths comparable to the large (bi)polaron's diameter. The resulting scattering time and mobility are inversely proportional to the thermal energy κT when it exceeds the energies of these phonons.

Doping is frequently used to introduce charge carriers to the high-dielectric-constant ionic insulators that are candidates for supporting large-bipolaronic superconductivity: e.g. SrTiO_3 and La_2CuO_4 . However, the dopant density need not be directly related to the density of mobile carriers. In some cases the multiplicity of valences that constituents of these ionic materials can assume (e.g. Tl can be 1+ or 3+) introduces uncertainty in the relationship between the doping density and the carrier density. Furthermore, some carriers may join solidified phases in which carriers order as small or large polarons or bipolarons in a manner which is commensurate with a crystal's underlying lattice structure. Then only the residual non-solidified carriers can contribute to a large-bipolaronic liquid.

A large-bipolaronic liquid and its superconductivity will disappear when carriers order in a manner that is commensurate with the solid's underlying lattice (Emin, 1994a; 1994b). If paired carriers form a square super-lattice whose period is d times the underlying lattice constant, commensurate arrangements will occur at carrier concentrations equaling the simple fractions $2/d^2$. As illustrated in Fig. 14.4, this situation may occur in $\text{A}_x\text{La}_{2-x}\text{CuO}_4$, where holes are introduced to the quasi-square CuO_2 planes of La_2CuO_4 by substituting divalent cations $\text{A} \equiv \text{Ca}, \text{Sr}$ or Ba for the trivalent cation La . In particular, the proximity of the compositional limits of the superconducting phase of $\text{A}_x\text{La}_{2-x}\text{CuO}_4$ to $x = 2/25$ and $x = 2/9$ with the loss of

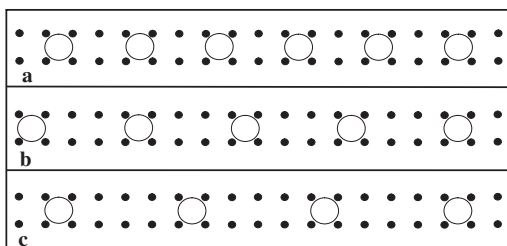


Fig. 14.4 Charge carriers in doped La_2CuO_4 are envisioned as residing in CuO_2 planes. Each plane is represented as a square array of Cu^{2+} cations (solid dots). To simplify this figure, oxygen O^{2-} anions, centered midway between each pair of nearest-neighbor Cu cations, are *not* shown. The core of each large bipolaron (open circles) is envisioned as encompassing the four contiguous oxygen anions bounded by the four Cu^{2+} cations of a square. At critical densities such large bipolarons can order commensurately with the underlying square lattice. In panels *a*, *b*, and *c* such ordering is depicted along one of the square lattice's principal axes when a large bipolaron is centered in every $3 \times 3 = 9$, $4 \times 4 = 16$ and $5 \times 5 = 25$ structural units, respectively.

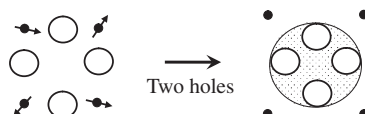


Fig. 14.5 An eight-atom structural unit of a CuO_2 plane of an insulating cuprate contains four Cu^{2+} cations (solid dots) each with a net spin of $\frac{1}{2}$ (arrow) and four contiguous O^{2-} anions (open circles). The shifting of cation and anion nuclei induced by adding two holes to the four-oxygen complex (dotted region), the core of the envisioned large-bipolaron, is hypothesized to transfer electrons from oxygen sites to copper sites leaving them with paired spins.

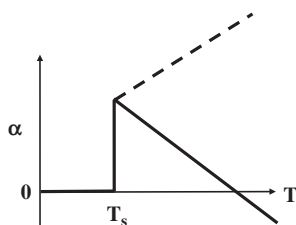


Fig. 14.6 The Seebeck coefficient α of a superconductor vanishes when the temperature T is lowered below the superconducting transition temperature T_s . When $T > T_s$ the Seebeck coefficient of a conventional p -type superconductor rises with increasing temperature (dashed line). If, however, large-bipolaron formation eliminates local magnetic moments, the Seebeck coefficient of a p -type cuprate superconductor may fall and even change sign as the temperature is raised (solid line).

superconductivity at the intermediate concentration $x = 2/16$ is consistent with the self-trapped carriers being paired.

The insulating parents of the cuprate superconductors, e.g. La_2CuO_4 , are anti-ferromagnets with spin- $\frac{1}{2}$ Cu^{2+} cations. With progressive doping, cuprates' magnetism disappears. If creating self-trapped holes eliminates local moments, carriers' Seebeck coefficients garner the negative spin entropy contribution described in Section 12.4. The magnitude of this negative contribution to the Seebeck coefficient increases with rising temperature as the removed-moments' spin-alignment entropy increases. For definiteness, consider the hypothesis illustrated in Fig. 14.5 in which formation of a p -type singlet large bipolaron centered on four contiguous oxygen ions removes the local moments of four adjacent spin- $\frac{1}{2}$ Cu^{2+} ions. This singlet-bipolaron formation process is described by the chemical formula: $2 \text{ holes} + (\text{O}_4)^{8-} + 4 \text{ Cu}^{2+} \rightarrow (\text{O}_4)^{2-} + 4 \text{ Cu}^{1+}$. Since four sites with spin- $\frac{1}{2}$ are eliminated in this process, it will contribute $-4(\kappa/2e)\ln(2)$ to the Seebeck coefficient in the high-temperature paramagnetic limit. This negative contribution to the Seebeck coefficient can drive it to resemble the schematic depiction of Fig. 14.6.

The superconducting condensate of a conventional metal will permeate into an adjoining metal thereby producing a proximity effect. Such proximity effects are not expected of a large-bipolaronic condensate since conventional metals will not sustain (bi)polaron formation. However, a large-bipolarons' condensate could extend into or pass through an electronic insulator whose static and optical dielectric constants support large-bipolaron formation, $\epsilon_0 \gg \epsilon_\infty$. Thereby a large-bipolaronic superconductor could exhibit novel proximity effects.